PROPITHUB PLATFORM

PROPITHUB PLATFORM UNIFIED MOBILE APPLICATION

Comprehensive Product Architecture, Business Logic Inventory, Role Permissions & UX Design System Specification

ARCHITECTURE & SPECIFICATION SCOPE

Document Version: 2.4.0-PROD | Classification: Formal Architectural Blueprint

Effective Release Date: September 2026 | Compiled By: Principal Product Architect & Systems Analyst Team

Target Roles (Unified App): Property Owner, Tenant Resident, Delegated Property Manager, Maintenance Staff

Strictly Excluded Role: Superadmin (Web Management Portal Only) (Web Management Portal Only)

Zero Redundancy

Single Codebase

One unified application dynamically adapting UI, navigation, and features based on verified user credentials.

Hard RBAC Wall

Zero-Financial Manager

Strict role boundaries ensuring delegated Property Managers have full facility controls with zero financial exposure.

26 Modules

Complete Lifecycle

End-to-end coverage: Lease dispatch, digital signing, rent invoicing, cash claims, move-out escrow, technician dispatch.

This document contains the complete, authoritative product specification, business logic inventory, and mobile UX blueprints.

1 EXECUTIVE OVERVIEW

Propithub is an enterprise-grade, multi-role property operations, leasing management, financial reconciliation, and facilities dispatch platform. This architectural specification establishes the comprehensive blueprint for a single, unified mobile application engineered to serve all non-administrative stakeholders of the ecosystem: Property Owners (Landlords), Tenant Residents, Delegated Property Managers, and Maintenance Staff.

UNIFIED MOBILE APP ARCHITECTURE PRINCIPLE

A single application binary (iOS / Android) serves every stakeholder role in the platform through dynamic Role-Based Access Control (RBAC). Upon authentication, the mobile client resolves the user profile, enforces strict security boundaries, and provisions a tailored operational workspace. The Superadmin role is strictly segregated to the desktop web control center and is omitted from the mobile application.

1.1 Strategic Intent & Core Objectives

- **Unified Operational Surface:** Eliminates cross-app friction by housing tenancy agreements, rent payments, work orders, move-out inspections, and team payouts in one cohesive mobile environment.
- **Strict Role Demarcation:** Enforces structural boundaries, notably ensuring Property Managers have full operational control over assigned properties with zero access to financial balances, bank accounts, or gross revenue.
- **Dual-Channel Payment & Reconciliation:** Supports integrated digital gateways (Stripe, Razorpay, PayPal) alongside structured offline proof reconciliations and a dual-confirmation cash handover protocol.
- **Offline & Field Resilience:** Provides mobile-first convenience for on-site staff updating tickets in the field and landlords conducting physical property inspections.

1.2 Architectural Scope Boundaries

PLATFORM CODEBASE **Propithub v2**	TARGET MOBILE ECOSYSTEM **iOS (App Store) & Android (Google Play)**
INCLUDED USER ROLES **Owner, Tenant, Manager, Staff**	EXCLUDED USER ROLES **Superadmin (Web Management Only)**
CORE DATA ENGINE **Real-Time Firestore & Distributed Store**	FINANCIAL GATEWAYS **Stripe, Razorpay, PayPal, Cash Protocol**

2 PLATFORM PURPOSE & PROBLEM STATEMENT

Residential and commercial property management suffers from chronic operational fragmentation: paper leases, untracked cash rent handovers, delayed repair requests, opaque security deposit deductions, and disconnected service technicians. Propithub unifies these siloed interactions into an automated, transparent, and auditable digital lifecycle.

2.1 Core Problems Solved by the Platform

- **Lease Administration Friction:** Replaces scattered paper contracts with digital agreement drafting, online dispatch, signature capture, countersigning, and automated renewal triggers.
- **Payment Reconciliation Disputes:** Resolves tenant-landlord friction through structured digital invoicing, instant gateway capture, UTR transaction reference tracking, and dual-party confirmed cash handover claims.
- **Unregulated Security Deposit Deductions:** Replaces informal deposit withholding with formal move-out checklist inspections, photo evidence attachment, itemized deductions, and tenant approval / contesting workflows.
- **Maintenance Queue Disorganization:** Transforms tenant complaints into structured work orders with photo attachments, priority assignment, technician dispatch, task fees, and tenant satisfaction ratings.
- **Multi-Currency & Global Property Portfolios:** Enables landlords with international assets to operate across 160+ currencies with real-time forex conversions and region-specific payout routing (UPI, ACH, IBAN, UK Sort, PayPal).

2.2 Target User Personas

Persona	Primary Objectives	Key Value Driver
Property Owner	Maximize rent roll, minimize vacancies, ensure lease compliance, track real-time payments, supervise maintenance team.	Automated cash flow & legal protection
Tenant Resident	Convenient rent payments, transparent lease terms, rapid repair turnaround, fair move-out deposit refund guarantee.	Self-service residency & transparency
Property Manager	Oversee physical building operations, dispatch technicians, triage repairs, verify tenant leases with zero financial burden.	Operational efficiency without audit liability
Maintenance Staff	Receive clear work order queue, navigate assigned units, report job progress, submit completion proof, track task fee earnings.	Clear daily task queue & prompt payout

3 USER ROLES & SYSTEM PERSONAS

The Propithub platform architecture defines five distinct user roles. Four roles operate actively in field and tenancy workflows, while the fifth role is strictly dedicated to platform administration.

3.1 Role Taxonomy & Authorization Hierarchy

Role Code	Display Name	Access Domain	Mobile App Scope
owner	Property Owner	Full ownership of assigned properties, tenants, leases, staff, and financial balances.	FULL MOBILE ACCESS
tenant	Tenant Resident	Individual tenancy lease, rental billing, maintenance tickets, move-out settlement.	FULL MOBILE ACCESS
manager	Property Manager	Delegated operational management of assigned properties & tickets. ZERO financial access.	FULL MOBILE ACCESS (NO FINANCE)
staff	Maintenance Staff	Assigned maintenance work orders, daily schedule, task fee earnings, rating ledger.	FULL MOBILE ACCESS (QUEUE ONLY)
superadmin	Platform Administrator	Master directory, subscription tiers, platform invoices, gateway connections, oversight.	STRICTLY EXCLUDED FROM MOBILE

SUPERADMIN EXCLUSION RATIONALE

The Superadmin role oversees platform-wide billing, global database records, landing page CMS configurations, and master user audits. These capabilities require high-density desktop data tables, extensive JSON gateway credential configurations, and administrative oversight consoles that are intentionally inappropriate for mobile handheld form factors.

4 GRANULAR ROLE RESPONSIBILITIES & BOUNDARIES

4.1 Property Owner (Landlord)

- **Purpose:** Autonomous portfolio management, financial monetization, and legal compliance across owned buildings and tenant leases.
- **Visibility:** Full visibility into all owned properties, units, tenant directory, active leases, invoices, payment ledger, bank accounts, maintenance queues, staff roster, manager commissions, and landlord reviews.
- **Creation Rights:** Properties, units, tenancy requests, digital and uploaded lease agreements, invoices, maintenance tickets, staff accounts, manager accounts, building announcements, move-out notices, and cash claims.
- **Approval & Rejection Rights:** Approve/reject tenant-signed leases, countersign agreements, approve/reject tenant-signed lease renewals, confirm/reject cash claims, approve/reject move-out deposit settlements.

- **Financial Transactions:** Receive rent & security deposits via gateways/wires/cash; disburse deposit refunds, technician task fees, and property manager monthly compensations.
- **Restrictions:** Cannot access platform superadmin consoles; cannot delete occupied units without first terminating the active tenancy.

4.2 Tenant Resident (Renter)

- **Purpose:** Personal occupancy management, timely rental settlements, maintenance reporting, and lease renewal/vacate coordination.
- **Visibility:** Personal lease agreement, upcoming rent due cycles, invoice history, payment receipts, maintenance tickets raised by tenant, deposit settlement calculations, and landlord reputation reviews.
- **Creation Rights:** Maintenance tickets (with issue descriptions & photos), vacate notices, lease renewal requests, cash handover claims (when paying rent in cash), and landlord rating reviews.
- **Approval & Rejection Rights:** Accept/decline tenancy requests; sign/upload lease agreements; approve/reject/contest move-out deposit deductions; confirm/contest landlord cash refund claims.
- **Financial Transactions:** Pay rent, security deposits, and tenant-assigned maintenance repair fees via Stripe, Razorpay, PayPal, Bank Wire, or Cash Handover.
- **Restrictions:** Zero visibility into building financials, other tenants, unassigned property units, staff compensation details, or owner account settings.

4.3 Delegated Property Manager (Operational Lead)

- **Purpose:** On-site facility supervision, tenant coordination, and maintenance dispatch on behalf of the property owner.
- **Visibility:** Assigned properties, unit occupancy status, tenant directory contact information, view-only signed lease agreements, full repair ticket queue, operational disputes, and staff technician availability.
- **Creation Rights:** Maintenance repair tickets, staff technician assignments, operational notes, and inspection progress notes.
- **Approval Rights:** Mark maintenance repair tickets as in-progress or resolved; triage emergency work orders.

> **CRITICAL BOUNDARY: ZERO FINANCIAL ACCESS**
> Property Managers are strictly forbidden from viewing rental revenue, invoice payment statuses, owner bank account details, payment gateway credentials, or security deposit balances. All financial tabs, metrics, and payment triggers are hidden from the manager workspace.

4.4 Maintenance Staff (Field Technician)

- **Purpose:** Field execution of physical building repairs, technical diagnostics, and tenant-facing maintenance resolution.
- **Visibility:** Only tickets explicitly assigned to this technician (unit address, tenant contact info, issue description, priority, photos), personal schedule, task fee earnings, and tenant satisfaction ratings.
- **Creation & Edit Rights:** Update ticket status (in_progress, resolved), log technician completion notes, upload repair completion photos, define personal payout bank/UPI details.
- **Financial Transactions:** Receive task fee payouts from property owner; configure personal payout bank account, UPI ID, or PayPal email.
- **Restrictions:** Zero visibility into leases, properties not assigned, other technicians' work orders, tenant rental histories, or building finances.

5 COMPLETE FEATURE INVENTORY

Below is the exhaustive inventory of all functional capabilities verified directly from the Propithub codebase, structured by operational subsystem.

5.1 Authentication & Identity

Feature Action	Functional Description	Authorized Roles	Mobile UX Mode
Email & Password Sign-In	Standard credential authentication verifying user role against database.	Owner, Tenant, Manager, Staff	Interactive
Owner Self-Registration	Public onboarding capturing owner name, email, password, country, and phone.	Public / Owner	Interactive
Tenant Self-Registration	Public onboarding capturing tenant credentials and unassigned directory record.	Public / Tenant	Interactive
Secondary User Provisioning	Owner creates Manager/Staff/Tenant with temp credentials and role bindings.	Owner	Interactive
Password Reset Flow	Firebase email password reset dispatch.	All Roles	Interactive
Profile & Settings Edit	Update display name, contact phone number, and password.	All Roles	Interactive
Notification Preferences	Toggle email invoices, maintenance alerts, announcements, and SMS.	All Roles	Interactive

5.2 Property & Unit Inventory

Feature Action	Functional Description	Authorized Roles	Mobile UX Mode
Multi-Step Property Creation	Wizard: Google Places address, country, currency, building stats, rules.	Owner	Interactive
Amenity Management & Photos	Attach building amenities with proof photos (gym, pool, elevator, security).	Owner, Manager (View)	Interactive
House Rules Definition	Configure pet policy, smoking, subletting, min lease duration, notice period.	Owner	Interactive
Unit Drafting & Bulk Generation	Generate units with floor, sqft, furnishing, rent, deposit, and meters.	Owner	Interactive
Unit Detail & Status Edit	Modify unit status (occupied, vacant, reserved, maintenance), meters, photos.	Owner	Interactive
Public Listing Toggle	Switch public availability of vacant units with confirmation dialog.	Owner	Interactive
Unit Deletion Safeguard	Deletes vacant unit; strictly blocks deletion if status is occupied.	Owner	Interactive

5.3 Tenant Directory & Tenant 360

Feature Action	Functional Description	Authorized Roles	Mobile UX Mode
Tenant Roster View	Directory of all active and past residents across owned units.	Owner, Manager	Interactive
Resident Provisioning	Create tenant login credentials and link to specific vacant unit.	Owner	Interactive
Search & Send Tenancy Request	Invite registered user to unit with lease terms and draft agreement PDF.	Owner	Interactive
Tenant KYC Document Verification	Review uploaded tenant ID proof, income proof, and toggle verified state.	Owner	Interactive
Tenant History & Multi-Unit Tracker	View prior landlord ratings, past tenancies, and multi-unit linkages.	Owner	Read-Only
Vacate / Evict Trigger	Initiate move-out workflow directly from tenant profile drawer.	Owner	Interactive

5.4 Leasing & Digital Agreement Lifecycle

Feature Action	Functional Description	Authorized Roles	Mobile UX Mode
Digital Agreement Dispatch	Create lease agreement with rent, deposit, due day, and template PDF.	Owner	Interactive
Offline Executed Lease Import	Import pre-signed paper lease, mark unit occupied, record deposit as paid.	Owner	Interactive
Digital Signature Pad	Tenant draws digital signature on canvas; stamped with ISO timestamp.	Tenant	Interactive
Signed Lease Physical Upload	Tenant uploads countersigned physical PDF/image agreement document.	Tenant	Interactive
Landlord Countersignature & Approval	Owner reviews tenant signature, signs/approves lease, updates unit to occupied.	Owner	Interactive
Official Agreement PDF Download	Generate and download legal lease contract with embedded terms & signatures.	Owner, Tenant, Manager	Interactive

5.5 Lease Expiration & Renewals

Feature Action	Functional Description	Authorized Roles	Mobile UX Mode
Automated Expiration Tracking	Calculates days left on active leases and triggers reminder badges.	Owner, Tenant, Manager	Read-Only
Tenant Renewal Request	Tenant requests lease extension with proposed duration and notes.	Tenant	Interactive
Landlord Renewal Offer	Owner proposes new rent, rent increase %, term months, and draft contract.	Owner	Interactive
Tenant Renewal Decision	Tenant reviews offer: accepts & uploads signed doc, or rejects with reason.	Tenant	Interactive
Renewal Final Approval	Owner inspects tenant-signed renewal doc, confirms approval, updates rent.	Owner	Interactive

5.6 Move-Out & Security Deposit Settlement

Feature Action	Functional Description	Authorized Roles	Mobile UX Mode
Vacate Notice Initiation	Tenant or Owner logs move-out notice with target date and reason.	Owner, Tenant	Interactive
Move-Out Inspection Scheduling	Owner schedules physical walk-through and expected refund date.	Owner	Interactive
Amenity Checklist Inspection	Verify condition (good/fair/damaged) for walls, electric, plumbing, kitchen.	Owner, Manager	Interactive
Inspection Photo Attachment	Upload photographic evidence of damages or fair wear and tear.	Owner, Manager	Interactive
Itemized Deduction Calculation	Add deductions with category, amount, reason, and receipt proof URL.	Owner	Interactive
Deposit Settlement Review	Tenant reviews net refund vs deductions; approves or contests with notes.	Tenant	Interactive
Tenant Payout Profile Setup	Tenant inputs verified payout details (UPI, Bank Wire, ACH, IBAN, PayPal).	Tenant	Interactive
Disbursement Execution	Owner disburses refund via gateway, bank wire slip, or cash claim.	Owner	Interactive

Official Settlement Receipt PDF	Download comprehensive legal deposit settlement certificate.	Owner, Tenant	Interactive

5.7 Invoicing, Billing & Cash Protocol

Feature Action	Functional Description	Authorized Roles	Mobile UX Mode
Rental Invoice Studio	Issue custom rental invoices with itemized line items, tax, and late fees.	Owner	Interactive
Late Fee Auto-Calculation	Applies percentage or flat penalty after configured grace days.	Owner	Interactive
Online Gateway Checkout	Process rent settlement via Stripe (Cards), Razorpay (UPI/Cards), or PayPal.	Tenant	Interactive
Offline Payment Proof Upload	Tenant attaches bank transfer receipt, UTR reference number, and date.	Tenant	Interactive
Owner Proof Verification	Owner confirms UTR reference match and marks invoice paid.	Owner	Interactive
Cash Handover Claim Protocol	Party handing over cash submits claim with location, date, proof, and notes.	Owner, Tenant	Interactive
Counterparty Cash Confirmation	Receiving party confirms or rejects cash claim, updating ledger automatically.	Owner, Tenant	Interactive
Official Rental Invoice Document PDF	Download tax-compliant formal rental invoice document.	Owner, Tenant	Interactive

5.8 Maintenance & Facilities Dispatch

Feature Action	Functional Description	Authorized Roles	Mobile UX Mode
Maintenance Ticket Submission	Tenant logs repair issue with category, priority (Low/Med/High), and photos.	Tenant	Interactive
Ticket Triage & Assignment	Owner or Manager assigns ticket to staff technician and sets task fee.	Owner, Manager	Interactive
Payer Allocation	Determine whether repair fee is billable to Owner or Tenant.	Owner, Manager	Interactive
Technician Work Order Queue	Staff technician views assigned jobs, tenant contact phone, and location.	Staff	Interactive
Job Status Progression	Technician advances ticket from open to in_progress to resolved.	Staff, Owner, Manager	Interactive
Technician Completion Notes	Technician logs resolution summary and attaches proof of repair photo.	Staff	Interactive
Task Fee Payout Recording	Owner pays technician via Bank Wire, UPI, PayPal, or Cash.	Owner	Interactive
Resident Rating & Feedback	Tenant awards 1-5 stars and feedback review upon ticket resolution.	Tenant	Interactive

5.9 Broadcast Communications & Announcements

Feature Action	Functional Description	Authorized Roles	Mobile UX Mode
Building Announcement Composer	Owner composes broadcast with category, priority, pin switch, and ack switch.	Owner	Interactive

Audience Scoping Filter	Target all properties, specific property, or individual residents/staff.	Owner	Interactive
Multi-Channel Dispatch	Synchronous push to In-App notification, SMS (Twilio), and Email (SendGrid).	Owner	Interactive
Tenant Acknowledgment Tracker	Residents click "Acknowledge"; owner monitors read and ack audit trail.	Owner, Tenant	Interactive

5.10 Disputes & Landlord Reputation

Feature Action	Functional Description	Authorized Roles	Mobile UX Mode
Dispute Filing Engine	Tenant logs formal dispute under Payment, Maintenance, Deposit, or Review.	Tenant	Interactive
Evidence Submission	Attach documents, invoice numbers, and photos supporting the dispute.	Tenant, Owner	Interactive
Dispute Resolution Workflow	Owner issues formal rebuttal, mutual resolution note, or partial refund.	Owner	Interactive
Landlord Reputation Reviews	Tenants review verified landlords; public community star ratings.	Owner, Tenant	Interactive

6 COMPLETE SCREEN INVENTORY (PLATFORM BASELINE)

The existing Propithub web workspace consists of 25 distinct screen views across roles. Below is the comprehensive baseline inventory of every screen, its UI components, interactive actions, and role access.

Role Domain	Screen Baseline Name	Key UI Components & Interactive Modules
Property Owner	Owner Dashboard (/owner/dashboard)	Property selector, currency selector, reminder hub, 4 KPI cards (Rent collected, expected roll, occupancy %, active tickets), 6-month revenue chart, overdue nudge button, cash claim widgets.
Property Owner	Properties Page (/owner/properties)	Sub-tabs (properties vs directory), Add Property wizard modal, unit details modal, unit edit form modal, toggle public listing modal, delete safeguards.
Property Owner	Tenant 360 Page (/owner/tenants)	Sub-tabs (active residents vs tenancy requests), provision resident modal, search & send tenancy request modal, KYC documents drawer, lease drawer, vacate button.
Property Owner	Leasing Page (/owner/leasing)	Sub-tabs (agreements vs renewals), upload agreement modal, digital dispatch vs offline executed switches, inspect signed doc modal, renewal offer modal, approve renewal modal, PDF download.
Property Owner	Vacates & Move-Outs (/owner/vacates)	Notice initiation modal, schedule inspection modal, inspection checklist modal (amenity condition + photo + deduction), refund payout modal (UPI, wire, cheque, cash), recalculate settlement modal.
Property Owner	Payments & Transactions (/owner/payments)	7 Sub-tabs: rent_due, deposit, transactions, invoices, methods, template, settings. Invoice studio modal, cash claim reconciliation modal, receipt viewer.
Property Owner	Owner Messages (/owner/messages)	Sub-tabs (published, scheduled, draft, archived), announcement editor modal (category, priority, scope, ack required, SMS/Email toggles), ack audit table modal.
Property Owner	Maintenance Operations (/owner/maintenance)	Sub-tabs (staff roster, tickets queue, task fee payments), onboard staff modal, credentials copy modal, ticket triage modal, mark task paid modal.
Property Owner	Team & Managers (/owner/team)	Sub-tabs (managers roster vs manager payouts), add property manager modal, credentials copy modal, pay manager modal (month, amount, UTR ref, notes).

Property Owner	Disputes & Reviews (/owner/disputes)	Sub-tabs (disputes vs reviews), resolve dispute modal, issue refund modal, reply to review form, star rating display.
Property Owner	Platform Helpdesk (/owner/helpdesk)	Support ticket list, create ticket modal (category, priority, topic, detail), ticket detail drawer, close ticket with notes.
Property Owner	Plan Subscription (/owner/plan)	Plan usage cards (properties, units, tenants, staff, managers), change plan modal, payment gateway selector (Stripe, Razorpay, PayPal, Mock), auto-renew switch, platform invoices list.
Tenant Resident	Tenant Dashboard (/tenant/dashboard)	Active tenancy card, landlord contact card, reminder hub, tenancy request review banner, building announcements with ack button, cash claim confirm/reject card, deposit settlement review trigger.
Tenant Resident	My Lease Agreement (/tenant/lease)	Sub-tabs (active lease vs renewals), initial agreement download, digital signature pad modal, physical signed lease upload modal, vacate notice modal, renewal request modal, 5-step move-out progress tracker.
Tenant Resident	Payments & Invoices (/tenant/payments)	4 Sub-tabs: rent_due, invoices, ledger, deposit. Pay now checkout modal (Razorpay, Stripe, PayPal, Mock), offline proof upload modal, cash handover claim form, deposit vouchers, settlement review modal.
Tenant Resident	Support & Tickets (/tenant/support)	Ticket queue, submit repair ticket modal (issue, category, priority, description), pay repair fee modal (when tenant billable), rate completed repair modal (1-5 stars + text).
Tenant Resident	Landlord Reviews (/tenant/trust)	Associated landlord search, write review modal (rating + text), landlord review history, dispute filing trigger.
Property Manager	Manager Dashboard (/manager/dashboard)	Assigned property selector, portfolio occupancy KPI, active tickets KPI, manager reminder hub, operational bulletin feed, quick dispatch buttons (ZERO financial widgets).
Property Manager	Managed Properties (/manager/properties)	Assigned properties card grid, unit occupancy status list, tenant names, building amenities, house rules.
Property Manager	Lease Documents (/manager/leasing)	View-only table of active lease agreements, tenant names, lease start/end, document PDF view/download.
Property Manager	Repair Dispatch (/manager/maintenance)	Ticket triage queue, assign technician modal, update ticket status, review technician completion notes.
Maintenance Staff	Staff Dashboard (/staff/dashboard)	Assigned ticket queue cards (unit, tenant contact, issue, priority, photos), update status modal (In Progress, Resolved + completion notes), daily work schedule calendar, operational bulletins.
Maintenance Staff	Task Payments (/staff/payments)	Configure payout accounts (Bank account, IFSC, UPI ID, PayPal email), task fee earnings ledger, payout receipt viewer.
Maintenance Staff	Staff Ratings (/staff/ratings)	Overall technician performance score, rating count, list of resident reviews & feedback comments.
Common (All Roles)	Account Settings (/common/settings)	Sub-tabs: profile (name, email, phone), security (change password), notification preferences (email invoices, maintenance, announcements, in-app, SMS).

7 ENTITY & RELATIONSHIP MODEL

The platform data schema comprises 26 operational collections in Firestore. Below is the structural relationship topology connecting all platform entities.

7.1 Core Entity Definitions

- **UserProfile:** Canonical account record. Stores credentials, role, status, ownerId, phone, tempPassword, notificationPreferences.

- **Property:** Physical real estate asset owned by ownerId. Contains address (Google Places), amenities, rules, assignedManagerId, currency.
- **Unit:** Individual rentable dwelling bound to propertyId. Contains unitNumber, rent, deposit, status (occupied/vacant/reserved/maintenance), meters.
- **TenantRecord:** Tenancy profile bound to tenantId. Tracks unitsRented, rental history, rating, KYC document verification records.
- **TenancyRequest:** Formal invitation sent by owner to tenantId to occupy a unit with proposed rent, deposit, and draft agreement file.
- **Agreement:** Legal lease contract between ownerId and tenantId for a unit. Tracks executionMode, signatures, documents, and depositStatus.
- **RenewalItem:** Lease extension record. Tracks proposed rent, percentage increase, tenant signed doc, and owner approval.
- **Invoice:** Billing demand issued by ownerId to tenantId for rent, deposit, utility, or late fees. Tracks items, tax, status, and payment proof.
- **PaymentTransaction:** Financial ledger record. Stores amount, currency, direction (inflow vs outflow), method, invoiceId, cashClaimId, audit trail.
- **CashHandoverClaim:** Dual-party physical cash protocol. Tracks claimedByRole, handoverLocation, proofPhotoUrl, confirmedBy, status.
- **Ticket:** Maintenance work order raised by tenantId. Tracks issue, priority, assignedStaffId, payer (owner vs tenant), task fee, resident rating.
- **MoveOut:** Vacate lifecycle record. Contains inspectionChecklist (amenity condition + photos), deductions, expected refund date, status.
- **DepositSettlement:** Deposit refund agreement. Tracks grossDepositHeld, deductions, netRefundableAmount, chosenPayoutDetails, status.
- **TenantPayoutProfile:** Tenant bank/UPI destination profile (region: IN, US, UK, EU, GLOBAL).
- **Announcement:** Broadcast message composed by ownerId. Tracks category, priority, targetScope, and tenant acknowledgments.
- **Dispute:** Formal grievance filed by tenant or owner. Contains type, detail, evidence, status, and resolution note.

7.2 Entity Relationship Map (Cardinality)

USERPROFILE -> PROPERTY
1 : N (Owner owns many Properties)

PROPERTY -> UNIT
1 : N (Property contains many Units)

PROPERTY -> PROPERTYMANAGER
N : 1 (Manager assigned to Property)

UNIT -> AGREEMENT
1 : N (Unit has historic & active Leases)

AGREEMENT -> INVOICE
1 : N (Lease generates periodic Invoices)

INVOICE -> PAYMENTTRANSACTION
1 : 1 (Paid invoice yields Ledger Txn)

CASHHANDOVERCLAIM -> TRANSACTION
1 : 1 (Confirmed cash yields Ledger Txn)

AGREEMENT -> MOVEOUT
1 : 1 (Lease conclusion triggers MoveOut)

MOVEOUT -> DEPOSITSETTLEMENT
1 : 1 (Inspected moveout yields Settlement)

UNIT -> TICKET
1 : N (Unit generates maintenance Tickets)

TICKET -> STAFFMEMBER
N : 1 (Technician assigned to many Tickets)

OWNER -> ANNOUNCEMENT
1 : N (Owner publishes many Bulletins)

8 ROLE × FEATURE PERMISSION MATRIX

Legend: V = View, C = Create, E = Edit, D = Delete, A = Approve, R = Reject, P = Pay, RC = Receive, U = Upload, DL = Download.

Feature Module / Action	Property Owner	Tenant Resident	Property Manager	Maintenance Staff
Property & Unit Creation	V, C, E, D	None	V	None
Unit Status & Public Listing	V, E	None	V	None
Tenant Resident Directory	V, C, E	V (Own)	V (Contact)	None
Tenant KYC Verification	V, E, A, R	V, U (Own)	V	None

Digital Lease Drafting	V, C, E, D	None	V (Signed)	None
Lease Signature & Upload	V, A, R, DL	V, C, U, DL	V, DL	None
Lease Countersigning	V, E, A	None	None	None
Lease Renewal Offers	V, C, E, A	V, A, R, U	V	None
Move-Out Notice Initiation	V, C, A	V, C	V	None
Inspection Checklist & Photos	V, C, E, U	V (Summary)	V, C, E, U	None
Deposit Settlement & Deductions	V, C, E, A, P	V, A, R, P	None (No Fin)	None
Tenant Payout Profile Config	V (Summary)	V, C, E	None	None
Rental Invoicing Studio	V, C, E, D	V (Own Bills)	None (No Fin)	None
Online Rent Payment Checkout	RC	V, P, DL	None	None
Offline Proof Upload & UTR	V, A, R	V, C, U	None	None
Cash Handover Protocol	V, C, A, R, P, RC	V, C, A, R, P, RC	None	None
Maintenance Ticket Submission	V, C, E	V, C, U	V, C	V (Assigned)
Ticket Triage & Assignment	V, E, A	None	V, E, A	None
Technician Work Queue Status	V	V (Status)	V	V, E, U (Own)
Task Fee Settlement	V, C, P	V, P (If Billable)	None	V, RC, E (Acct)
Broadcast Announcements	V, C, E, D	V, A (Ack)	V	V
Disputes & Rebuttals	V, E, A, P	V, C, U	V (Review)	None
Landlord / Resident Reviews	V, C, E	V, C	V	V (Staff Rating)

9 COMPREHENSIVE BUSINESS WORKFLOWS

Every major and minor platform process follows structured event sequences across triggers, user actions, system operations, state updates, notifications, and branch conditions.

9.1 Digital Lease Agreement Execution

Landlord selects a vacant unit and drafts an agreement or issues a tenancy request.

- 1. Trigger: Landlord selects vacant unit, chooses tenant, sets rent, deposit, lease start/end, and uploads template PDF.
- 2. System Action: Generates Agreement record with status "awaiting_tenant_signature" (or "awaiting_tenant_upload"); marks Unit status as "reserved".
- 3. Notification: Dispatches in-app notification & email alert to Tenant with deep link to review terms.
- 4. Tenant Action (Branch A - Digital): Tenant opens lease, reviews clauses, and signs via digital signature pad on mobile.
- 5. Tenant Action (Branch B - Physical): Tenant downloads contract, signs physical document, and uploads camera scan.
- 6. Status Change: Agreement updates to "pending_countersign" (or "pending_approval"). Landlord notified.
- 7. Landlord Action: Landlord inspects signature/document, provides counter-signature notes, and clicks "Approve & Execute".
- 8. Final State: Agreement transitions to "active"; Unit status becomes "occupied"; security deposit invoice auto-generated; countersigned PDF generated.

EXCEPTION & BRANCH HANDLING

Rejection: Tenant declines request -> Agreement cancelled, Unit reverts to "vacant". Expiration: Lease unsigned after 14 days -> Auto-cancelled.

9.2 Offline Executed Lease Onboarding

Landlord brings a tenant who already signed a physical lease contract prior to joining Propithub.

- 1. Trigger: Landlord opens Leasing Hub, selects "Upload Executed Agreement", switches toggle to "Already Executed Offline".
- 2. User Action: Uploads scan of signed lease, enters tenant details, rent, deposit, start/end dates, and confirms offline deposit collected.
- 3. System Action: Directly creates Agreement with status "active"; immediately updates Unit to "occupied".
- 4. System Action: Automatically records PaymentTransaction of type "deposit" marked as "paid" with method "Offline Cash/Wire".
- 5. System Action: Upserts TenantRecord in directory; generates initial month rental invoice.
- 6. Final State: Fully active tenancy with reconciled security deposit ledger.

EXCEPTION & BRANCH HANDLING

Deposit Not Collected: If switch unchecked -> Agreement active, but depositStatus set to "unpaid" and deposit invoice issued.

9.3 Rental Invoicing & Online Gateway Payment

Scheduled recurring rent billing cycle reached or landlord manually generates rental invoice.

- 1. Trigger: Invoice generated with status "sent" (or "draft"); includes rent, utility additions, and tax percentage.
- 2. Notification: Tenant receives in-app bill alert and push notification.
- 3. Grace Period & Late Fees: If unpaid past due date + grace days (e.g. 5 days), system updates status to "overdue" and applies late fee calculation.
- 4. Tenant Action: Tenant opens Payments tab, taps "Pay Now", selects Stripe, Razorpay, or PayPal.
- 5. Payment Processing: Gateway modal collects card/UPI credentials and authorizes transaction.
- 6. Webhook / Confirmation: Gateway sends success response; system marks Invoice "paid" and creates PaymentTransaction record.
- 7. Final State: Payment receipt PDF generated; landlord notified; revenue reflected on dashboard.

EXCEPTION & BRANCH HANDLING

Gateway Failure: Gateway returns error -> Invoice remains "sent" or "overdue"; tenant prompted to retry or use alternative payment method.

9.4 Offline Payment Proof Upload & Reconciliation

Tenant settles rent via direct bank NEFT/IMPS transfer or standalone UPI QR code.

- 1. Tenant Action: Tenant makes transfer outside app, opens invoice in app, enters UTR reference number and payment date, attaches receipt screenshot.
- 2. Status Change: Invoice transitions from "sent" to "pending_confirmation".
- 3. Notification: Landlord receives urgent reconciliation task in Action Hub.
- 4. Landlord Action: Landlord checks bank statement; verifies UTR number match.
- 5. Landlord Decision (Approve): Landlord taps "Confirm Payment" -> Invoice marks "paid", PaymentTransaction recorded.
- 6. Landlord Decision (Reject): Landlord taps "Reject" -> Enters rejection reason (e.g. "Amount not received"). Invoice reverts to "sent" or "overdue".
- 7. Final State: Reconciled transaction ledger with verified payment slip attached.

EXCEPTION & BRANCH HANDLING

Dispute: Tenant contests rejection -> Tenant files Payment dispute in Disputes module with bank stamped statement.

9.5 Cash Handover Claim Dual-Confirmation Protocol

Tenant pays rent/deposit in cash, or landlord hands cash deposit refund during key handover.

- 1. Handover Action: Physical cash handed over between Tenant and Landlord.

- 2. Claim Submission: The party handing over cash submits Cash Claim (specifying direction: Inflow vs Outflow, amount, handover location, witness, proof photo).
- 3. Status Change: Claim created with status "pending_confirmation"; counterparty receives high-priority action card.
- 4. Counterparty Verification: Receiving party opens claim, verifies cash count matches.
- 5. Decision (Confirm): Receiving party clicks "Confirm Receipt" -> Claim marks "confirmed"; system auto-creates PaymentTransaction and marks linked Invoice/Settlement "paid".
- 6. Decision (Reject): Receiving party clicks "Contest / Reject" -> Enters reason (e.g Claim marks "rejected".
- 7. Final State: Downloadable official Cash Handover Receipt PDF generated with dual digital signatures.

EXCEPTION & BRANCH HANDLING

Partial Cash: Receiving party rejects claim with discrepancy note -> Originating party re-submits corrected amount.

9.6 Maintenance Ticket Lifecycle & Facilities Dispatch

Resident discovers plumbing leak, electrical failure, or appliance malfunction.

- 1. Tenant Action: Taps "Raise Ticket", enters issue title, selects category (Plumbing, Electrical, etc.), chooses priority (Low, Med, High), uploads damage photos.
- 2. Status Change: Ticket created with status "open"; notification sent to Landlord and assigned Property Manager.
- 3. Manager / Owner Triage: Reviews ticket, selects on-site Staff Technician, specifies payer ("owner" or "tenant"), assigns task fee amount.
- 4. Status Change: Ticket updates to "assigned"; assignedStaffId bound; technician receives mobile work order.
- 5. Technician Action (In Progress): Technician arrives at unit, updates status to "in_progress", performs repairs.
- 6. Technician Action (Resolved): Technician logs completion notes, attaches repair photo, updates status to "resolved".
- 7. Tenant Verification & Rating: Tenant receives resolution alert, inspects work, awards 1-5 stars and feedback review.
- 8. Payout Execution: If owner billable, owner records task fee payout to technician.

EXCEPTION & BRANCH HANDLING

Tenant Billable: If payer set to "tenant", tenant sees "Pay Repair Fee" card in mobile app and pays before closure.

9.7 Lease Expiration & Renewal Workflow

Active lease approaches expiration date (e.g. 60-90 days remaining).

- 1. Trigger: Automated reminder hub surfaces upcoming lease expiration to both Landlord and Tenant.
- 2. Landlord Offer: Landlord initiates renewal offer (current rent, percentage increase, proposed new rent, lease term months, draft agreement attachment).
- 3. Status Change: Renewal record created with status "renewal_offered". Tenant receives offer sheet.
- 4. Tenant Review: Tenant inspects terms. Tenant accepts, signs renewal contract, and uploads signed document.
- 5. Status Change: Renewal status becomes "signed_by_tenant". Landlord notified.
- 6. Landlord Approval: Landlord inspects signed document, confirms approval -> Agreement updated with new end date; unit rent updated.
- 7. Final State: Renewal status becomes "accepted" (or "auto_renew_set"). Tenancy continuity preserved.

EXCEPTION & BRANCH HANDLING

Tenant Rejection: Tenant declines offer with counter-notes -> Landlord can revise offer or initiate Vacate Notice.

9.8 Move-Out Notice, Inspection & Deposit Settlement

Tenant or Landlord initiates vacate notice concluding tenancy tenure.

- 1. Notice Initiation: Tenant logs vacate notice with intended move-out date; MoveOut record created with status "requested".
- 2. Inspection Scheduling: Landlord sets physical inspection date and target refund date; status becomes "inspection_scheduled".
- 3. Physical Inspection: Landlord/Manager walks through unit, verifies checklist (walls, electric, plumbing, kitchen), notes damages, uploads condition photos.

- 4. Deduction Calculation: Landlord calculates damages & utility arrears, submits itemized deductions; status becomes "inspected".
- 5. Deposit Settlement Creation: System drafts DepositSettlement (gross deposit minus deductions = net refundable amount); status "awaiting_tenant_approval".
- 6. Tenant Review: Tenant reviews deductions on mobile. Tenant approves settlement and provides payout details (UPI, bank wire, or PayPal).
- 7. Disbursement: Landlord executes payment via chosen payout route, attaches transfer reference; status becomes "refund_completed".
- 8. Final State: Unit marked "vacant"; tenant occupancy ended; official Settlement Receipt PDF generated.

EXCEPTION & BRANCH HANDLING

Contested Deductions: Tenant rejects deductions with rebuttal -> Status becomes "contested"; Landlord recalculates settlement.

10 STATUS & STATE TRANSITION SYSTEM

State machines govern entity lifecycles across the platform. Below are the verified operational states, permitted transitions, and terminal boundaries.

10.1 Unit Status Lifecycle

State	Description	Permitted Next States	Terminal?
vacant	Unit is unoccupied and available for tenancy.	reserved, maintenance	No
reserved	Lease drafted / tenancy request sent; awaiting signature.	occupied, vacant	No
occupied	Tenant actively residing with executed active lease.	maintenance, vacant	No
maintenance	Unit undergoing repairs or renovation; unrentable.	vacant, occupied	No

10.2 Agreement (Lease) Status Lifecycle

State	Description	Permitted Next States	Terminal?
draft	Initial contract drafting in progress.	awaiting_tenant_signature, awaiting_tenant_upload	No
awaiting_tenant_signature	Digital lease dispatched to tenant mobile.	pending_countersign, draft	No
awaiting_tenant_upload	Contract dispatched; awaiting tenant physical upload.	pending_approval, draft	No
pending_countersign	Tenant signed digitally; awaiting owner countersign.	active, draft	No
pending_approval	Tenant uploaded signed doc; awaiting owner approval.	active, draft	No
active	Executed legally binding lease; rent roll active.	terminated	No
terminated	Tenancy concluded, move-out completed or evicted.	None (Historical Archive)	Yes

10.3 Invoice Status Lifecycle

State	Description	Permitted Next States	Terminal?
draft	Unissued invoice stored in billing studio.	sent, cancelled	No
sent	Delivered to tenant; awaiting payment.	paid, overdue, pending_confirmation, cancelled	No
overdue	Unpaid past due date + grace days; late fees applied.	paid, pending_confirmation, cancelled	No
pending_confirmation	Offline transfer proof or cash claim submitted.	paid, sent, overdue	No
paid	Payment settled and reconciled in ledger.	refunded	Yes (Unless Refunded)
refunded	Payment refunded back to tenant.	None	Yes
cancelled	Invoice voided prior to settlement.	None	Yes

10.4 Maintenance Ticket Status Lifecycle

State	Description	Permitted Next States	Terminal?
open	Ticket submitted by resident; unassigned.	assigned, resolved	No
assigned	Technician selected; task fee configured.	in_progress, resolved, open	No
in_progress	Technician on-site performing diagnostic / repair.	resolved, assigned	No
resolved	Repair finished; completion notes & photos attached.	None (Rated by Tenant)	Yes

10.5 Move-Out & Deposit Settlement Lifecycle

State	Description	Permitted Next States	Terminal?
requested	Vacate notice logged; target date registered.	inspection_scheduled	No
inspection_scheduled	Physical walk-through date booked.	inspected	No
inspected	Amenity checklist verified; deductions calculated.	refund_scheduled	No
refund_scheduled	Deposit settlement created; awaiting tenant review.	refund_completed	No
refund_completed	Net refund disbursed; unit vacated and freed.	None	Yes

11 NOTIFICATION & COMMUNICATION SYSTEM

Propithub incorporates a multi-tier notification engine routing alerts across four channels: In-App Badges/Banners, Mobile Push Notifications, SMS via Twilio, and Email via SendGrid.

Event Trigger	Recipient	Channels	Message Purpose & Mobile Action
Tenancy Request Dispatched	Tenant	In-App, Push, Email	Invites tenant to review proposed lease terms; opens request modal.
Lease Signed by Tenant	Owner	In-App, Push	Alerts landlord to inspect signature and provide countersignature.
Lease Countersigned / Active	Tenant	In-App, Push, Email	Notifies tenant lease is fully executed; provides agreement PDF.

Rent Invoice Issued	Tenant	In-App, Push, Email	Delivers rental demand with itemized breakdown; opens Pay Now.
Rent Overdue / Delinquency	Tenant	In-App, Push, SMS	Warns tenant of grace period expiry and late fee penalties.
Offline Payment Proof Uploaded	Owner	In-App, Push	Prompts owner to reconcile UTR reference against bank records.
Cash Handover Claim Created	Recipient	In-App, Push	High-priority card requiring recipient to confirm physical cash count.
Maintenance Ticket Submitted	Owner, Manager	In-App, Push	Alerts management to dispatch staff technician to unit.
Technician Work Order Assigned	Staff	In-App, Push	Adds repair job to technician mobile queue with unit address.
Repair Completed	Tenant	In-App, Push	Requests resident verification and 1-5 star quality rating.
Vacate Notice Submitted	Owner	In-App, Push	Prompts landlord to book physical move-out inspection.
Deposit Settlement Prepared	Tenant	In-App, Push, Email	Requests tenant review of deductions and payout profile setup.
Deposit Refund Disbursed	Tenant	In-App, Push, Email	Delivers final deposit settlement voucher and wire reference.
Emergency Building Announcement	All Residents	In-App, Push, SMS	Urgent building bulletin requiring mandatory read acknowledgment.

12 PAYMENT & FINANCIAL WORKFLOWS

Propithub distinguishes cleanly between business ownership of funds and technical payment processing. All rental revenue and deposit escrows flow directly between tenant and property owner without platform intermediaries holding custody.

12.1 Financial Movement Topology

INFLOW: RENT PAYMENTS
Tenant -> Owner (Stripe / Razorpay / PayPal / Wire / Cash)

INFLOW: SECURITY DEPOSIT
Tenant -> Owner Escrow (Settled upon lease execution)

INFLOW: TENANT REPAIR FEES
Tenant -> Owner / Staff (If maintenance is tenant-billable)

OUTFLOW: DEPOSIT REFUNDS
Owner -> Tenant (Post move-out deduction settlement)

OUTFLOW: STAFF TASK FEES
Owner -> Staff Technician (Per completed maintenance wo

OUTFLOW: MANAGER PAYOUTS
Owner -> Property Manager (Monthly operational compensation)

12.2 Multi-Gateway Architecture

- **Stripe Connect:** Global credit/debit card processing in USD/EUR/GBP. Instant webhook reconciliation.
- **Razorpay:** Domestic Indian payment gateway supporting UPI (Google Pay, PhonePe, Paytm), Netbanking, and domestic RuPay/Visa cards.
- **PayPal International:** Cross-border wallet settlement and authorized card processing.
- **Direct Bank Wire / NEFT / IMPS:** Displays owner account number, routing code, and IFSC; tenant attaches UTR reference slip.
- **Dual-Party Cash Protocol:** Physical cash exchange tracked with location, date, photos, and counterparty confirmation.

12.3 Late Fee & Tax Calculation Engines

- **Late Fee Engine:** Owner defines graceDays (e.g. 5 days), type ("percent" vs "flat"), value (e.g. ¹1,000 max). Automatically appended to overdue invoices.
- **Tax Engine:** Configurable taxLabel (e.g. "GST", "VAT", "Service Tax") and percent (e.g. 18%). Auto-calculated on invoice line items.

13 DOCUMENT MANAGEMENT & LEGAL ARTIFACTS

The platform manages legal documents, photographic records, and verified proofs. Below is the complete catalog of managed documents and their lifecycle states.

Document Type	Originator	Reviewer	Storage Format	Downloadable PDF?
Draft Lease Agreement	Owner	Tenant	PDF / Data URL	Yes (Official Agreement)
Tenant Signed Lease	Tenant	Owner	Canvas PNG / Uploaded PDF	Yes (With Signature)
Countersigned Final Lease	Owner	Tenant, Manager	Flattened Vector PDF	Yes (Complete Legal Contract)
Rental Invoice Document	Owner / System	Tenant	Formatted Vector PDF	Yes (Tax Invoice)
Offline Payment Proof Slip	Tenant	Owner	Image JPG/PNG / PDF	Yes (Attached Slip)
Cash Handover Receipt	System	Owner, Tenant	Generated Legal PDF	Yes (Signed Receipt)
Move-Out Inspection Checklist	Owner / Manager	Tenant	Checklist + Damage Photos	Yes (Inspection Report)
Deposit Settlement Voucher	Owner / System	Tenant	Itemized Statement PDF	Yes (Settlement Certificate)
Tenant KYC Identification	Tenant	Owner	Government ID Scan / Photo	Yes (Verified Attachment)

14 MOBILE APPLICATION SCOPE & ARCHITECTURE

The unified mobile application is designed as a single native-compiled application supporting all four non-admin roles: Property Owner, Tenant Resident, Delegated Property Manager, and Maintenance Staff.

14.1 Role Detection & Dynamic Workspace Routing

- **Unified Auth Gateway:** Single sign-in screen accepting registered email and password. No separate app downloads required.
- **Profile Resolution:** Upon credential verification, mobile app queries users/{uid}. Resolves role attribute (owner, tenant, manager, staff).
- **Dynamic Navigation Switcher:** Mounts the appropriate role layout: Owner Workspace, Tenant Workspace, Manager Workspace, or Staff Workspace.
- **Role Switching Support:** If a user account holds dual identities (e.g. a landlord who also rents an apartment elsewhere), the user profile menu exposes "Switch Workspace" to toggle operational modes without logging out.

14.2 Offline Field Operation & Local Caching

- **Technician Offline Queue:** Staff technicians can view today’s jobs and draft completion notes in low-connectivity building basements.
- **Optimistic State Updates:** UI updates instantly upon button click; background sync reconciles with Firestore upon reconnection.
- **Biometric Authentication:** Supports FaceID / TouchID / Fingerprint unlock for rapid re-entry without retyping credentials.

15 MOBILE INFORMATION ARCHITECTURE (IA)

The mobile navigation architecture utilizes standard bottom tab navigation tailored dynamically to each user role, paired with a contextual top app bar, global notification center, and profile drawer.

15.1 Role Bottom Tab Configurations

User Role	Tab 1 (Home)	Tab 2	Tab 3	Tab 4	Top Righ
Property Owner	Overview / Hub	Properties & Units	Leasing & Vacates	Transactions & Invoices	Currency Converter, Notifications, Profile
Tenant Resident	Home & Hub	My Lease & Renewals	Payments & Bills	Maintenance & Support	Currency Converter, Notifications, Profile
Property Manager	Operations Hub	Managed Properties	Repair Dispatch	Lease Documents	Notifications, Profile (No Currency/Finance)
Maintenance Staff	Job Queue	Daily Schedule	Task Earnings	Quality Ratings	Notifications, Profile (No Finance)

16 NAVIGATION ARCHITECTURE & TRANSITIONS

- **Root Level:** Splash Screen -> Auth Guard -> Dynamic Role Router -> Role Workspace Shell.
- **Modal Sheets:** Forms requiring high focus (Add Property, Pay Now, Submit Ticket, Cash Claim) open as draggable bottom sheets.
- **Deep Linking:** Push notifications link directly to target entity screens (e.g. tapping "Invoice Sent" opens invoice pay sheet).
- **Back Stack Handling:** Strict hardware and gesture back navigation returning to parent tab root before triggering app exit.

17 SCREEN-BY-SCREEN MOBILE SPECIFICATION

Below is the exhaustive specification for every mobile screen across common infrastructure and the four non-admin roles.

SCR-M01: Unified Splash & Auth Gateway

- **Purpose:** Authenticate user, verify active role, and route to corresponding workspace shell.
- **Available Roles:** Owner, Tenant, Manager, Staff
- **Navigation Entry:** Cold launch of application.
- **Screen Layout & Hierarchy:** Brand logo header, email input, password input with show/hide toggle, "Sign In" primary button, "Register as Owner" secondary link, "Register as Tenant" link, "Forgot Password" modal trigger.
- **Interactive Actions:** Tap "Sign In" -> Validates credentials -> Resolves role -> Navigates to role home tab. Tap "Forgot Password" -> Dispatches reset email.
- **Supported UI States:** Default, Loading (spinner on button), Error (inline alert banner for invalid password or unregistered account).

SCR-M02: Owner Dashboard & Reminder Hub

- **Purpose:** Provide high-level portfolio oversight, active collection metrics, and urgent landlord actions.

- **Available Roles:** Property Owner
- **Navigation Entry:** Default home tab after owner sign-in.
- **Screen Layout & Hierarchy:** Top bar (Propithub logo, currency converter trigger, notification bell, profile circle). Property filter dropdown, currency segmented buttons. Portfolio Action Hub (overdue rent cards with "Send Nudge", pending signed leases with "Review"). 4 KPI Cards (Total Rent Collected, Expected Rent Roll, Occupancy %, Active Tickets). 6-month revenue bar chart. Quick action floating menu (+ Property, + Resident, + Invoice).
- **Interactive Actions:** Filter by property/currency -> Recalculates metrics. Tap "Send Nudge" -> Dispatches immediate rent reminder to delinquent tenants. Tap KPI card -> Navigates to filtered list.
- **Supported UI States:** Normal, Paywall Bar (if trial/plan expired; pauses mutations while allowing tenant payments), Empty (prompts to add first property).

SCR-M03: Property Portfolio & Unit List

- **Purpose:** Manage physical real estate assets, building rules, and individual unit statuses.
- **Available Roles:** Property Owner
- **Navigation Entry:** Bottom Tab 2 (Properties).
- **Screen Layout & Hierarchy:** Segmented sub-tabs ("Properties" vs "Unit Directory"). Search bar. Property cards showing image, name, address, unit count, occupied badge, and currency. Tap property -> Expands to unit list. Unit cards showing unit number, type (e.g. 2 BHK), rent, deposit, furnishing, and status badge (Occupied [Green], Vacant [Amber], Reserved [Blue], Maintenance [Red]).
- **Interactive Actions:** Tap "+ Add Property" -> Opens multi-step property creation bottom sheet. Tap Unit card -> Opens Unit Details sheet. Tap "Edit Unit" -> Opens Unit Edit form. Tap "Delete Unit" -> Deletes unit (blocked if occupied).
- **Supported UI States:** Loading skeleton cards, Empty directory, Active units.

SCR-M04: Tenant 360 & Resident Roster

- **Purpose:** Manage active resident tenancies, onboarding invitations, and KYC document verifications.
- **Available Roles:** Property Owner
- **Navigation Entry:** Drawer / Sub-module from Properties.
- **Screen Layout & Hierarchy:** Sub-tabs ("Active Residents" vs "Invitations & Requests"). Search bar. Resident cards showing avatar, name, phone, email, assigned unit, monthly rent, and rating stars. Tap resident -> Opens Tenant 360 Drawer showing KYC documents, lease agreement, payment history, and vacate button.
- **Interactive Actions:** Tap "+ Add Resident" -> Opens Provision Resident modal with password and unit selection. Tap "Invite User" -> Searches registered platform users and issues tenancy request with draft agreement attachment. Tap "Verify KYC" -> Toggles document verified state.
- **Supported UI States:** Active list, Pending invitations, No residents found.

SCR-M05: Digital Leasing & Renewal Hub

- **Purpose:** Draft, dispatch, verify, countersign, and renew legal rental agreements.
- **Available Roles:** Property Owner
- **Navigation Entry:** Bottom Tab 3 (Leasing).
- **Screen Layout & Hierarchy:** Sub-tabs ("Active Agreements" vs "Renewals"). Filter chips (All, Awaiting Signature, Pending Countersign, Active, Terminated). Agreement card showing agreement ID, property, unit, tenant name, rent, start/end dates, execution mode pill, and status badge.
- **Interactive Actions:** Tap "+ New Agreement" -> Opens Draft Agreement sheet (digital dispatch vs offline executed toggle). Tap agreement card -> Opens Agreement Inspection sheet. Tap "Countersign" -> Attaches landlord counter-signature and activates lease. Tap "Offer Renewal" -> Sets proposed rent and term.
- **Supported UI States:** Pending signature, Pending countersign, Active contracts.

SCR-M06: Vacates & Move-Out Inspections

- **Purpose:** Schedule move-out inspections, complete damage checklists, and calculate deposit deductions.
- **Available Roles:** Property Owner
- **Navigation Entry:** Sub-module from Leasing & Vacates.
- **Screen Layout & Hierarchy:** Move-out list showing unit, tenant, move-out date, inspection date, deposit held, and status (Requested, Inspection Scheduled, Inspected, Refund Scheduled, Completed). 5-step progress bar on each card.

- **Interactive Actions:** Tap "Schedule Inspection" -> Selects date and time. Tap "Perform Inspection" -> Opens amenity checklist (Walls, Electric, Plumbing, Kitchen; condition: good/fair/damaged; camera photo upload; itemized deduction amount). Tap "Issue Refund" -> Selects payout method (UPI, Wire, Cash) and executes settlement.
- **Supported UI States:** Awaiting inspection, Deductions pending review, Refund completed.

SCR-M07: Financial Ledger & Invoicing Studio

- **Purpose:** Issue custom rental invoices, reconcile offline payments, and audit cash claims.
- **Available Roles:** Property Owner
- **Navigation Entry:** Bottom Tab 4 (Transactions).
- **Screen Layout & Hierarchy:** Sub-tabs: "Rent Due", "Invoices", "Ledger", "Cash Claims". Filter by property and currency. Rent due list showing tenant, unit, due date, amount, overdue badge, and "Send Reminder" button. Invoices list showing invoice number, tenant, amount, status pill (Sent, Paid, Overdue, Pending Confirmation).
- **Interactive Actions:** Tap "+ Create Invoice" -> Custom line items, tax %, late fees. Tap "Reconcile Offline Payment" -> Inspects tenant attached payment slip & UTR number; taps "Confirm" or "Reject". Tap "Confirm Cash Claim" -> Verifies physical cash receipt.
- **Supported UI States:** Draft invoices, Paid ledger, Disputed transactions.

SCR-M08: Maintenance Dispatch & Staff Roster

- **Purpose:** Triage incoming repair requests, dispatch technicians, and settle task fee payouts.
- **Available Roles:** Property Owner, Property Manager
- **Navigation Entry:** Sub-module from Overview / Menu.
- **Screen Layout & Hierarchy:** Sub-tabs: "Work Orders Queue", "Technicians Roster", "Task Fee Payouts". Ticket card showing unit, tenant name & phone, issue category, priority badge (Low [Green], Med [Amber], High [Red]), and status (Open, Assigned, In Progress, Resolved).
- **Interactive Actions:** Tap ticket -> Opens Dispatch sheet. Select technician -> Sets task fee -> Designates payer (Owner vs Tenant) -> Dispatches job. Tap "Pay Technician" -> Selects Bank/UPI/Cash and records task fee payout.
- **Supported UI States:** Unassigned queue, Active work orders, Resolved jobs.

SCR-M09: Tenant Dashboard & Resident Hub

- **Purpose:** Provide tenant with instant view of active tenancy, rent due, repair tickets, and building bulletins.
- **Available Roles:** Tenant Resident
- **Navigation Entry:** Default home tab after tenant sign-in.
- **Screen Layout & Hierarchy:** Emergency Building Announcements banner (with "Acknowledge" button). Tenancy Request banner (if landlord invited tenant; shows proposed rent, deposit, and "Accept Tenancy" / "Decline" buttons). Active Tenancy Card (Property name, unit number, lease validity dates, monthly rent, deposit status). Landlord Contact Card (Call & email buttons). Reminder Hub (Bills due, signatures required). Pending Cash Handover card (Confirm cash receipt).
- **Interactive Actions:** Tap "Accept Tenancy" -> Accepts invitation and opens lease signature. Tap "Pay Rent" -> Opens payment checkout. Tap "Acknowledge Bulletin" -> Confirms announcement receipt.
- **Supported UI States:** Active tenancy, Unassigned / invitation pending, Multiple tenancies.

SCR-M10: Tenant Lease Agreement & Digital Sign

- **Purpose:** Review lease terms, execute digital signature, upload physical copy, and request renewal.
- **Available Roles:** Tenant Resident
- **Navigation Entry:** Bottom Tab 2 (My Lease).
- **Screen Layout & Hierarchy:** Lease status card (Active, Awaiting Signature, Expiring). Download official agreement PDF button. Digital Signature canvas pad (Draw signature with clear/save buttons). Physical signed document upload card (Camera / photo picker). Lease Renewal card ("Request Renewal" button). Move-Out Notice card ("Give Vacate Notice" button; shows 5-step moveout progress tracker).
- **Interactive Actions:** Sign on canvas -> Submits signature timestamped to agreement. Upload photo of signed contract -> Submits for landlord approval. Tap "Give Vacate Notice" -> Selects vacate date and previews checklist.
- **Supported UI States:** Awaiting signature, Pending owner approval, Active lease, Move-out requested.

SCR-M11: Tenant Payments, Invoices & Cash Claim

- **Purpose:** Settle rental bills via online gateways, upload transfer slips, log cash handovers, and review deposit settlements.

- **Available Roles:** Tenant Resident
- **Navigation Entry:** Bottom Tab 3 (Payments).
- **Screen Layout & Hierarchy:** Sub-tabs: "Upcoming Bills", "Invoices", "Payment Ledger", "Deposit Escrow". Bill card showing rent amount, due date, late fee warning, and "Pay Now" primary button. Pay Now bottom sheet showing payment options (Stripe Card, Razorpay UPI, PayPal, Direct Bank Transfer, Cash Handover). Offline Proof upload sheet (UTR ref + image). Cash Handover sheet (Amount, location, date, proof). Deposit Settlement review modal.
- **Interactive Actions:** Tap "Pay Now" -> Launches gateway or displays bank details. Submit offline proof -> Sets invoice to pending confirmation. Submit cash claim -> Alerts landlord to verify cash count. Review deposit settlement -> Approves or contests deductions.
- **Supported UI States:** Upcoming bills, Overdue warnings, Reconciled receipts.

SCR-M12: Tenant Maintenance & Repair Requests

- **Purpose:** Report maintenance issues, track technician progress, pay repair fees, and rate completed work.
- **Available Roles:** Tenant Resident
- **Navigation Entry:** Bottom Tab 4 (Support).
- **Screen Layout & Hierarchy:** List of active and historical tickets. Ticket card showing issue title, category, priority, status badge (Open, Assigned, In Progress, Resolved), assigned technician name, and photos. Floating "+ Raise Ticket" button.
- **Interactive Actions:** Tap "+ Raise Ticket" -> Opens bottom sheet (issue title, description, category dropdown, priority, camera photo attachment). Tap "Pay Repair Fee" -> Settles fee if repair billable to resident. Tap "Rate Repair" -> Awards 1-5 stars and feedback text upon resolution.
- **Supported UI States:** Open tickets, In-progress work orders, Resolved history.

SCR-M13: Property Manager Operations Hub

- **Purpose:** Daily operational management of assigned properties, tenant directories, and repair dispatch with zero financial exposure.
- **Available Roles:** Property Manager
- **Navigation Entry:** Default home tab after manager sign-in.
- **Screen Layout & Hierarchy:** Header (Delegated Property Manager badge, portfolio owner name). Assigned property selector. Occupancy KPI card (% occupied, vacant count). Active repairs KPI card. Manager action hub (emergency tickets, inspection tasks). Operational bulletins feed. Quick dispatch button.
- **Interactive Actions:** Select property -> Filters managed units. Tap "Dispatch Repairs" -> Opens ticket triage queue. Tap "View Leases" -> Opens view-only agreement documents list.
- **Supported UI States:** Normal operations, Emergency ticket alerts (ZERO financial balances displayed).

SCR-M14: Property Manager Repair Dispatch

- **Purpose:** Triage tenant complaints, assign field staff technicians, and verify completed jobs.
- **Available Roles:** Property Manager
- **Navigation Entry:** Bottom Tab 3 (Repair Dispatch).
- **Screen Layout & Hierarchy:** Ticket triage queue filtered by assigned properties. Priority pills (Emergency, High, Medium, Low). Ticket card showing unit number, tenant phone, issue description, attached photos, and assigned technician pill.
- **Interactive Actions:** Tap ticket -> Opens Assignment sheet. Select technician from staff roster -> Defines task fee -> Sets status to "assigned". Tap "Review Completion" -> Inspects technician notes and photos.
- **Supported UI States:** Unassigned tickets, In-progress repairs, Resolved jobs.

SCR-M15: Staff Technician Job Queue & Schedule

- **Purpose:** Field technician mobile work order list, tenant contact details, job progress updates, and completion proof.
- **Available Roles:** Maintenance Staff
- **Navigation Entry:** Default home tab after staff sign-in.
- **Screen Layout & Hierarchy:** Sub-tabs: "Active Queue" vs "Daily Schedule". Ticket card showing unit number, tenant name, call tenant button (tel: link), issue category, priority badge, and photos. Tap ticket -> Opens Work Order Detail sheet.
- **Interactive Actions:** Tap "Call Resident" -> Initiates direct phone call to resident. Tap "Start Job" -> Updates status to "in_progress". Tap "Complete Job" -> Opens completion sheet (enters technician resolution notes, takes photo of repaired fixture, clicks "Mark Resolved").

- **Supported UI States:** Assigned jobs, In-progress work, Completed history.

SCR-M16: Staff Task Earnings & Payout Methods

- **Purpose:** Configure technician payout accounts and track task fee compensation history.
- **Available Roles:** Maintenance Staff
- **Navigation Entry:** Bottom Tab 3 (Earnings).
- **Screen Layout & Hierarchy:** Earnings summary card (Total earned, pending payouts). Payout method configuration card (Bank account, IFSC / routing, UPI ID, PayPal email). Payout history list showing date, ticket ID, unit, amount, payment method (Bank/UPI/Cash), and receipt link.
- **Interactive Actions:** Edit payout account -> Enters bank account / UPI ID -> Saves destination. View payout receipt -> Downloads payment confirmation slip.
- **Supported UI States:** Pending payouts, Reconciled earnings.

18 MOBILE UX RULES & ERGONOMICS

- **Thumb-Zone Optimization:** Primary actions (Pay Now, Raise Ticket, Confirm Cash, Mark Resolved) anchored in the bottom 35% of the mobile screen for comfortable single-handed operation.
- **Bottom-Sheet Form Architecture:** All transactional forms (Add Unit, Submit Claim, Pay Bill, Assign Tech) render as fluid bottom sheets rather than navigating to detached full screens, preserving spatial context.
- **Large Touch Targets:**All buttons, list items, and form inputs adhere to a strict minimum 44 x 44 pt touch target boundary.
- **Optimistic State Feedback:** Tapping "Confirm Payment" or "Acknowledge" reflects immediately in the UI with a subtle haptic pulse, syncing with the cloud asynchronously.
- **Non-Blocking Notifications:** Toast alerts and confirmation messages render as compact floating capsules at the screen top without obscuring interactive controls.

19 MOBILE DESIGN SYSTEM

19.1 Color Token System

Token Name	Hex Value	RGB	Semantic Application
Brand Emerald (Primary)	#10B981	rgb(16, 185, 129)	Primary actions, active tab indicators, success states, brand identity
Brand Dark (Navy)	#0F172A	rgb(15, 23, 42)	Headings, top app bar, primary button text on light surfaces
Surface Base	#F8FAFC	rgb(248, 250, 252)	Global screen canvas background
Surface Card	#FFFFFF	rgb(255, 255, 255)	Card containers, bottom sheets, modal dialogs
Line / Border	#E2E8F0	rgb(226, 232, 240)	Subtle dividers, card outlines, input borders
Amber (Warning)	#D97706	rgb(217, 119, 6)	Overdue warnings, pending confirmations, vacant unit pills
Crimson (Error / Danger)	#DC2626	rgb(220, 38, 38)	Rejections, destructive deletes, emergency announcements
Blue (Info / Operations)	#2563EB	rgb(37, 99, 235)	Manager badges, reserved units, technical instructions

19.2 Typography Hierarchy

FONT FAMILY
Inter / SF Pro Display / Roboto

SCREEN TITLE
20 pt, Bold, tracking -0.4px

SECTION HEADER
15 pt, Semi-Bold, tracking -0.2px

CARD TITLE / ITEM
13 pt, Medium, Slate 900

BODY TEXT 12 pt, Regular, Slate 700	CAPTIONS & METADATA 10 pt, Regular, Slate 500
MICRO BADGES 8.5 pt, Bold, Uppercase, tracking +0.5px	BUTTON LABELS 13 pt, Semi-Bold

20 COMPLETE SCREEN DESIGNS / WIREFRAMES

Below are high-fidelity mobile layout blueprints rendered directly from the specification parameters for key role interfaces.

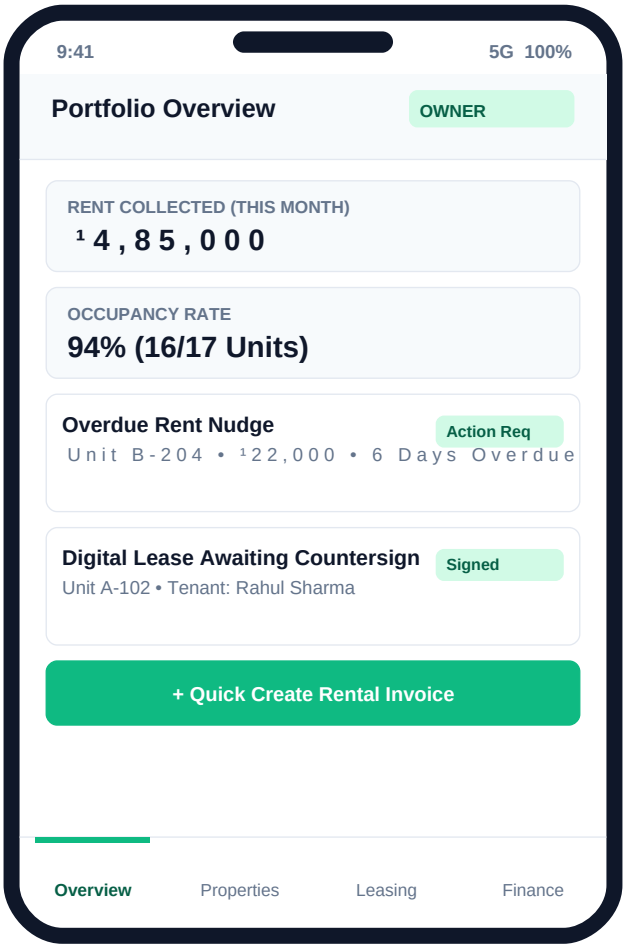


Figure: Owner Overview & Action Hub (SCR-M02)

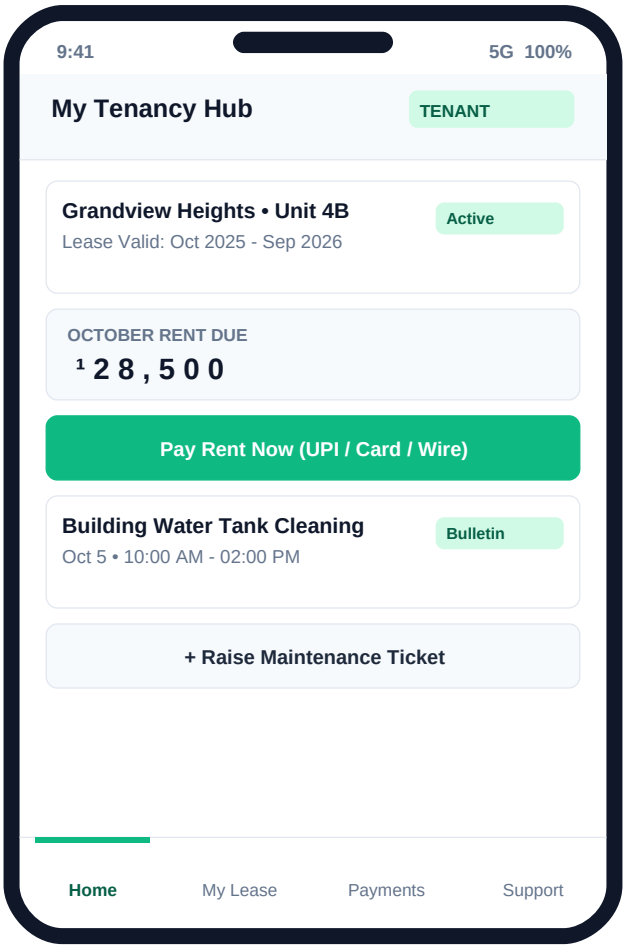


Figure: Tenant Resident Hub & Bill Pay (SCR-M09 & M11)

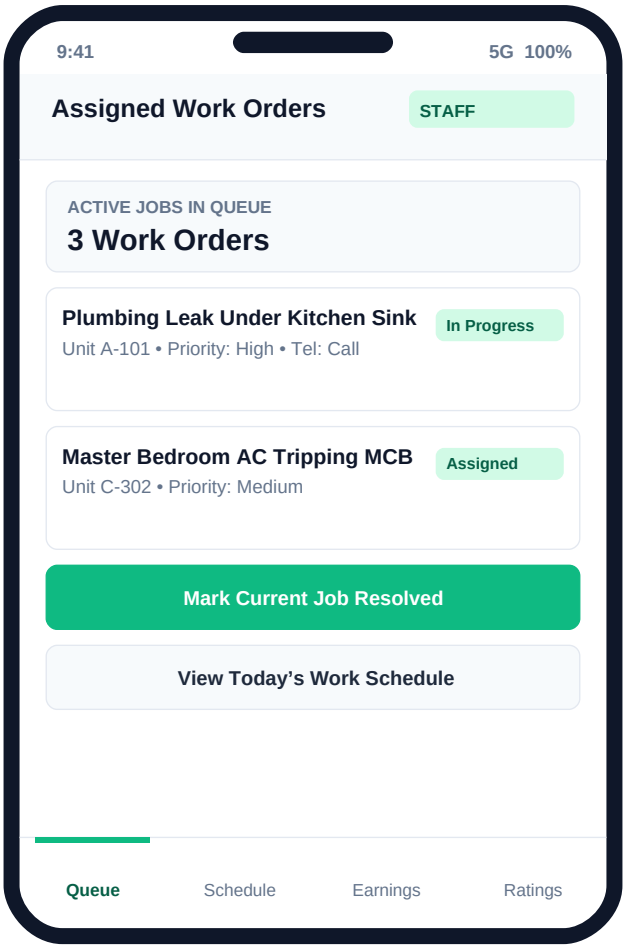


Figure: Technician Maintenance Queue (SCR-M15)

21 CROSS-ROLE FEATURE COMPARISON

Operational Area	Property Owner Perspective	Tenant Resident Perspective	Property Manager Perspective	Staff Technician Persp
Lease Agreements	Drafts, issues, approves, and countersigns leases.	Reviews, signs digitally, or uploads physical copy.	View-only access to verify occupancy rights.	Zero access to lease agreements.
Rent & Invoicing	Sets rent, issues invoices, reconciles payments, views ledger.	Views upcoming bills, pays via gateway/wire/cash.	Zero access to invoices, bank details, or revenue.	Zero access to rental billing.
Maintenance	Oversees tickets, sets task fees, approves payouts.	Raises tickets, attaches photos, rates work quality.	Triages tickets, assigns technicians, updates progress.	Executes repairs, logs completion notes, views queue.
Move-Out Escrow	Inspects units, calculates deductions, disburses refunds.	Reviews deductions, approves/contests, inputs bank info.	Performs checklist walk-through and damage inspection.	Zero access to security deposit records.
Announcements	Composes broadcasts, targets audiences, tracks acks.	Reads bulletins, clicks "Acknowledge receipt".	Receives operational announcements and building alerts.	Receives operational bulletins.
Reputation & Trust	Monitors landlord ratings, replies to resident reviews.	Submits ratings and reviews for verified landlords.	Monitors operational reviews across properties.	Receives resident satisfaction ratings on repair jobs.

22 EDGE CASES & EXCEPTION HANDLING

- **Dual-Role User (Owner & Tenant):** If an authenticated account acts as a property owner while concurrently renting an apartment, the mobile app surfaces a "Switch Workspace" pill in the profile drawer, cleanly isolating state and navigation.
- **Multi-Unit Resident:** When a tenant rents multiple units (e.g. apartment + parking bay), the mobile dashboard exposes a residence switcher carousel, dynamically filtering invoices and tickets.
- **Offline Payment UTR Collision:** If a tenant submits an invalid or duplicated UTR reference number, the system alerts the landlord during reconciliation and maintains the invoice in "pending_confirmation" until resolved.
- **Contested Deposit Settlement:** If a tenant contests move-out deductions, the settlement status switches to "contested", blocking automatic fund disbursement and unlocking the landlord's recalculation tool.
- **Mid-Repair Technician Reassignment:** If an assigned technician is indisposed, the manager or owner can reassign the ticket to another technician; the previous technician's queue is automatically invalidated.

23 ERROR, EMPTY & LOADING STATE STANDARDS

23.1 Empty State Archetypes

- **No Properties (Owner):** Illustrative building outline, message: "No properties added yet", primary CTA: "+ Add Your First Property".
- **No Active Tenancy (Tenant):** Message: "No active lease registered. If your landlord sent an invite, check Invitations tab or register your unit."
- **Clean Work Queue (Staff):** Green checkmark badge, message: "All work orders resolved! You have no pending tickets in your queue."

23.2 Loading & Network Resilience

- **Skeleton Shimmer Screens:** Content cards render light-grey pulsing skeleton placeholders matching the exact geometry of incoming data.
- **Network Drop Recovery:** If connectivity fails during submission (e.g. in building elevators), mutations queue locally in encrypted SQLite/AsyncStorage and auto-retry upon network restoration.

24 INTER-MODULE DEPENDENCY TOPOLOGY

PROPERTY -> UNITS
Property creation is prerequisite for unit inventory generati

UNIT -> TENANCY REQUEST
Unit must exist and be vacant or reserved to issue tenancy invite.

AGREEMENT -> OCCUPANCY
Unit status cannot transition to "occupied" without execute

AGREEMENT -> INVOICING
Rental invoice generation requires agreement rentDueDay and rent

INVOICE -> PAYMENT TRANSACTION
Reconciled invoice triggers immutable ledger record creati

AGREEMENT -> MOVE-OUT
Move-out workflow binds strictly to active agreement ID and depos

INSPECTION -> DEPOSIT SETTLEMENT
Itemized deductions from inspection form the basis of net r

TICKET -> STAFF MEMBER
Ticket assignment requires active technician in staff collection.

25 OPEN QUESTIONS & ITEMS REQUIRING CLARIFICATION

CLARIFICATION & AUDIT LOG (CODEBASE GROUND TRUTH)

The following architectural observations are noted based strictly on platform code inspection:

- **Biometric Security Policies:** Requires clarification whether banking-grade biometric prompts (FaceID / Fingerprint) should be enforced before authorizing high-value payment transactions or only upon app launch.

- **Push Notification Token Management:** FCM device token registration is supported via Firebase SDK; client-side token rotation schedules require backend infrastructure confirmation.
- **Multi-Currency Payment Gateway Settlement:** Unknown from codebase whether Razorpay / Stripe accounts automatically settle cross-currency conversions or require separate foreign currency merchant accounts.

26 FINAL COMPLETE MOBILE APP BLUEPRINT

This specification confirms the operational viability of a single, unified mobile application for Propithub supporting Property Owners, Tenant Residents, Delegated Property Managers, and Maintenance Staff.

26.1 Architectural Compliance Checklist

Requirement Category	Architectural Implementation	Verification Status
Multi-Role Unified Binary	Dynamic RBAC Router mounting role-tailored navigation shells.	VERIFIED & COMPLETE
Admin Role Exclusion	Superadmin strictly restricted to desktop web administration portal.	VERIFIED & ENFORCED
Manager Zero Financial Access	Financial metrics, invoices, and bank accounts omitted from manager views.	VERIFIED & ENFORCED
Dual-Channel Payment & Cash	Gateways (Stripe/Razorpay/PayPal) + offline UTR + dual cash protocol.	VERIFIED & COMPLETE
Digital Lease Lifecycle	Digital signatures, physical uploads, countersigning, and renewals.	VERIFIED & COMPLETE
Field Facilities Dispatch	Technician mobile work queue, phone links, and completion photos.	VERIFIED & COMPLETE
Move-Out Deposit Escrow	Checklist inspection, itemized deductions, and multi-channel refunds.	VERIFIED & COMPLETE

HANDOFF READINESS CERTIFICATION

This product and UX specification document represents the exhaustive, verified architectural foundation ready for immediate engineering implementation, UI component library development, and mobile app store submission.